

Rishav Chourasia

OBJECTIVE  +65-97168493 |  rishav@betterdata.ai |  [rishav1.github.io](https://github.com/rishav1) |  Google Scholar

Veteran AI/ML researcher with a Ph.D. and over 6 years of experience with privacy engineering. Expert in differential privacy, multiparty computation, and cryptography with published works in venues like FORC, TDP, and NeurIPS. Presently, my research focuses on Privacy-Preserving ML, Machine Unlearning, and Differential Privacy Auditing.

EXPERIENCE

- Betterdata.ai** Dec 2025 - Present & Aug 2023 - Nov 2025
Head of Product (Earlier: Research Scientist / Privacy Engineer) Singapore
 - Lead development of the IRG1 platform; won IEEE S&P Distinguished Paper Award for auditing Apple's proprietary DP framework running on billions of devices.
 - Authored 4 papers on differential privacy quantification and auditing, with accepts at IEEE S&P, FMSD @ ICML, PPAI @ AAAI, FORC, and TDP.
 - Coauthored grant proposal to secure [US Department of Homeland Security's S&T Awards Contract](#).
- Google Brain** May 2022 - Aug 2022
PhD Student Researcher with [Dr. Abhradeep Guha Thakurta](#) and [Dr. Krishnamurthy \(Dj\) Dvijotham](#) Singapore
 - Derived high-confidence bounds on predictions for differentially private uncertainty estimation in Lasso regression.
- Google Summer of Code** May 2019 - Aug 2019
Student Developer with [Preferred Networks](#) Remote
 - Contributed to a [Python-to-ONNX JIT compiler](#) for Chainer ML framework.
 - Contributions enabled [JIT compilation support for complex models](#) like LSTMs, AlexNet and ResNet50.
 - Esteemed [Shinichiro Hamaji](#) described my contributions as "*outstanding*".
- Max-Planck Institute of Software Systems** Nov 2018 - Feb 2019
Research Fellow with [Dr. Adish Singla](#) Saarbrücken, Germany
 - Studied how voting rules affect exploration/exploitation in multi-agent Q-learning, resulting in a NeurIPS paper.
- Amazon Fraud Detection Team** July 2018 - Nov 2018 & May 2017 - July 2017
Software Development Engineer & Software Development Intern Bangalore, India
 - Performance optimizations and bug fixes on the fraud detection ML platform (<800 ms latency with 99.9% SLA).
 - Developed tooling to track changes in pipeline configurations and raise alerts.

EDUCATION

- National University of Singapore** Aug 2019 - May 2025
Ph.D. in Computer Science with advisor [Dr. Xiaokui Xiao](#) Singapore
 - GPA: 4.83/5.00 - [PhD Thesis](#)
- Indian Institute of Technology Guwahati** July 2014 - June 2018
B.Tech in Computer Science and Engineering Guwahati, India
 - GPA: 9.26/10.00

SELECTED PROJECTS

- Privacy Auditing: Empirically Detecting Violations of Differential Privacy** March 2023 - Aug 2025
Keywords: [Differential Privacy, Private Federated Learning, Reconstruction Attacks] 
 - Reverse-engineered Apple's [DifferentialPrivacy.framework](#) to find DP bugs affecting 87% of audited macOS Sonoma collection; IEEE S&P Distinguished Paper Award.
 - Pioneered a new operational interpretation of differential privacy, simplifying privacy analysis. This work received an oral acceptance at [FORC](#), [TDP](#) and [PPAI](#), three of the most prestigious gatherings for privacy researchers.
- Machine Unlearning: How to Guarantee Deletion in ML** March 2022 - Feb 2023
Keywords: [Noisy Gradient Descent, Trustworthy Deletion] 
 - Built a [rigorous and sound notion](#) for quantifying data-deletion capacity of unlearning algorithms.
 - Proved that *unlearning via Noisy Gradient Descent* has optimal privacy-utility trade-offs with few training passes.
 - Results published at [ICML 2023](#) and at [FORC 2023](#) (20-minute oral presentation).
- Privacy-Preserving ML: Maximizing Utility with Minimal Compute** March 2020 - Feb 2022
Keywords: [Differential Privacy, Noisy Gradient Descent] 
 - Proved $O(1)$ [privacy bound for Noisy Gradient Descent](#), a result that is regarded as a "*breakthrough*".
 - Collaborated with NEC researchers to develop the [cross-distillation technique](#) to enhance privacy.
 - Research works published in [NeurIPS 2021](#) (spotlight) and [PETS 2021](#).
- Chainer Compiler: An experimental Python-to-ONNX converter** May 2019 - Aug 2019
Tools: [ONNX, Chainer ML]  
 - Developed [Elichika](#), a compiler to convert Python source of a [chainer.Chain](#) object into ONNX computation graph.
 - Added [core language features](#) to the compiler, such as support for lambda functions, looping, and return.
 - Contributions were featured in [NumFOCUS blog](#).

PUBLICATIONS

AUTHORSHIP ORDER BY CONTRIBUTION. C=CONFERENCE, W=WORKSHOP, T=THESIS

- [C.1] Rishav Chourasia, Ergute Bao, Uzair Javaid, and Xiaokui Xiao. (2026). **Auditing Apple's Differential Privacy framework: Implementation Bugs, Misconfigurations, and Practical Risks**. In *IEEE S&P 2026*, pp. 1815–1834. Distinguished Paper Award.
- [C.2] Rishav Chourasia, et al. (2025). **Laplace Transform Interpretation of Differential Privacy**. In *6th Symposium on Foundations of Responsible Computing (FORC 2025)* (pp 11:1–11:22).
- [C.3] Rishav Chourasia and Neil Shah. (2023). **Forget Unlearning: Towards True Data-Deletion in Machine Learning**. In *International Conference on Machine Learning (ICML 2023)* (pp. 6028–6073). PMLR.
- [C.4] Rishav Chourasia, et al. (2021). **Differential Privacy Dynamics of Langevin Diffusion and Noisy Gradient Descent**. In *Advances in Neural Information Processing Systems 34 (NeurIPS 2021)*, pp. 14771–14781.
- [C.5] Rishav Chourasia, et al. (2021). **Knowledge Cross-Distillation for Membership Privacy**. In *Proceedings on Privacy Enhancing Technologies (PETS 2021)*, pp 362–377.
- [C.6] Rishav Chourasia, et al. (2017). **Visualization of Two-Dimensional Interval Type-2 Fuzzy Membership Functions using General Type-2 Fuzzy Membership Functions**. In *17th World Congress of International Fuzzy Systems Association (IFSA-SCIS 2017)*, pp. 1–5.
- [C.7] V Saxena, N Yadala, Rishav Chourasia, et al. (2017). **Type Reduction Techniques for Two-Dimensional Interval Type-2 Fuzzy Sets**. In *International Conference on Fuzzy Systems (FUZZ-IEEE 2017)*, pp. 1–6.
- [W.1] Rishav Chourasia and Adish Singla (2019). **Unifying Ensemble Methods for Q-learning via Social Choice Theory**. In *Learning with Rich Experience workshop (LIRE 2019)* at NeurIPS 2019.
- [W.2] Rishav Chourasia, et al. (2025). **One-Run Privacy Auditing for Structured Generative and Foundation Models**. In the 1st ICML Workshop on Foundation Models for Structured Data, Vancouver, Canada. 2025.
- [T.1] Rishav Chourasia. (2025). **Learning and Unlearning via Noisy Gradient Descent**. Doctoral Dissertation, National University of Singapore.
- [T.2] Rishav Chourasia. (2018). **Optimal Swarm RL: An Improved Deep-Exploration Strategy**. Bachelor Dissertation, Indian Institute of Technology Guwahati.

SKILLS

- **Programming Languages:** Python, C, C++, Objective C, Java
- **Machine Learning Frameworks:** JAX, PyTorch, Keras, ONNX, Fastai, Scikit-learn, XGBoost, Chainer
- **Cloud Technologies:** AWS, GCP, Docker, CI/CD
- **Mathematical & Statistical Tools:** Mathematica, MATLAB, SymPy, NumPy, Jupyter, Pandas, Polars
- **Database Systems:** MySQL, PostgreSQL, SQLite, MongoDB
- **DevOps & Version Control:** Git, CircleCI, Github Actions, Jenkins
- **Specialized Area:** Performance Optimization, ML Theory, Trustworthy AI Fundamentals
- **Other Tools & Technologies:** Weights & Biases, Comet ML, FastAPI, Flask
- **Research Skills:** Problem Formulation, Critical Thinking, Data Collection, Grant Writing, Presentation

HONORS AND AWARDS

- **SoC Research Incentive Award** Oct 2023
National University of Singapore
◦ Presented to students who have demonstrated good academic standing and research progress. 
- **Research Achievement Award** June 2022
National University of Singapore
◦ Presented to PhD students who have achieved outstanding research performance over the past academic year. 
- **1st Prize, CS6101 Deep Unsupervised Learning course** Dec 2019
15th SoC Term Project Showcase, School of Computing
- **PhD Research Scholarship** May 2019
National University of Singapore
◦ Awarded to promising PhD candidates accepted at NUS; 100% tuition fee-waiver and stipend for four years.
- **Research Fellowship** Sept 2018
Max Planck Institute for Software Systems
◦ Invited as a visiting scholar to work on a research project in the area of machine teaching. 
- **Department Topper** July 2015
Indian Institute of Technology, Guwahati
◦ Achieved highest GPA among freshman students in my department and was among top 8/440 students that year.
- **High School Matriculation Topper** March 2012
DeNobili C.M.R.I, Dhanbad
◦ Achieved the highest percentage (96.4%) in ICSE matriculation among over 500 students.